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25 APRIL 1984
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9 MARCH 1982
AND AMENDMENT 1
10 JANUARY 1983

MILITARY SPECIFICATION

Closure Assembly, Aft, For Mk 56 Mod 2 and Mod 3 Rocket Motors

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers one type of aft closure assembly hereinafter referred to as the closure (see 6.6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

Specifications

Federal

O-M-232	Methanol (Methyl Alcohol)
TT-I-735	Isopropyl Alcohol

Standards

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

Beneficial comments (recommendations, additions, deletions), and pertinent data which may be of use in improving this document, should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (524), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-STD-410	Nondestructive Testing Personnel Qualification and Certification
DOD-STD-2101	Classification of Characteristics

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

Drawings

Naval Sea Systems Command (Code Ident 53711)

5546252	Closure Assembly, Aft

Naval Ordnance Systems Command (Code Ident 10001)

2841603	Shell-Closure, Aft

Publications

Naval Ordnance Systems Command (Code Ident 10001)

WS 4503	Rocket Motor Mark 56 Mod 2 and Mod 3

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 304-58	Normal Butyl Alcohol (Butanol)
ASTM D 494	Acetone Extraction of Phenolic Molded or Laminated Products
ASTM D 792	Specific Gravity and Density of Plastics by Displacement
ASTM D 2583	Indentation Hardness of Plastics by Means of a Barcol Impressor
ASTM E 114	Ultrasonic Pulse-Echo Straight-Beam Testing by the Contact Method

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Traffic Department, 1616 P Street, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1107, 222 South Riverside Plaza, Chicago, IL 60606.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified in the contract (see 6.2), first article samples of the closure are required and shall be manufactured using the methods and procedures of the manufacturing process specification (see 3.4.2). The samples shall be tested as specified in 4.4 for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing closures that comply with the technical requirements of the contract.

3.2 Classification of characteristics. The characteristics specified herein are classified as critical, major or minor in accordance with DOD-STD-2101. Critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). All other characteristics are classified minor. The number following the classification symbol is the serial number of the characteristic to be verified. The following exception to DOD-STD-2101 applies to classification of drawing characteristics:

When the classification applies to only a portion of a note or dimensional characteristic, or there are several classifications applicable, the applicable portions of the note shall be enclosed in brackets in lieu of parentheses, followed by the classification code and serial number.

3.3 Materials. All materials and parts shall be in accordance with Dwg 5546252.

3.4 Design and fabrication. The closure design and fabrication shall be in accordance with Dwg 5546252 and the following;

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3.4.1 Bonding (M101). The find nos. 1 and 3 details of Dwg 5546252 shall be bonded to the primed steel shell (Dwg 2841603) using sufficient adhesive to provide a minimum bonded area of 75% of the area in each of the 1.65 inch cylindrical section and in the 2.90 inch conical section of the aft closure as defined in FIGURE 1. Bondline thickness shall be in accordance with Dwg 5546252. No individual void, unbond or porous area (see 6.6.5) within the above defined conical and cylindrical sections shall exceed 0.5 square inches. The bondline of each cylindrical and conical section shall be circumferentially continuous with a width not less than 0.5 inch.

* 3.4.2 Manufacturing process specification. Unless otherwise specified in the contract (see 6.2), the contractor shall prepare a manufacturing process specification. The process specification shall fully describe all manufacturing and fabricating processes and methods of control of manufacturing variables, including proper storage and use of adhesives and primer, bondline thickness control, control of adhesives and primer application, prefitting of parts, cleaning of parts, handling of cleaned parts, control of prepregs, curing of adhesive and primer, etc. (see 6.7). All processing shall be in accordance with the requirements of the processing specification.

3.4.3 Molding. Molding of the molded components (see 6.6.2) shall be at a minimum of 2000 pounds per square inch (psi) plus an additional 500 psi on the molding material per each inch of component height. Positive compression molding (no stops) shall be used for first article inspection samples (see 4.4) and all production molding of components.

3.5 Chemical and physical properties. The chemical and physical properties of components molded in accordance with 3.4.3 shall be as specified in TABLE I.

TABLE I. Chemical and physical properties.

Molded Component	Acetone Extractable, % Maximum	Specific Gravity, Minimum	Barcol Hardness Minimum
-1	0.75	1.72	67
-3	1.00	1.35	55

3.6 Performance. The closure assembly shall meet the following requirements after firing with a MK 56 Mod 2 or Mod 3 rocket motor.

3.6.1 Erosion (M102). The throat diameter at the point of maximum restriction shall not exceed 2.92 inches after firing. Gouges or discontinuously eroded areas shall be less than 0.13 inch deep or 0.50 inch wide. The minimum remaining thickness shall be not less than that specified for the points indicated in FIGURE 2.

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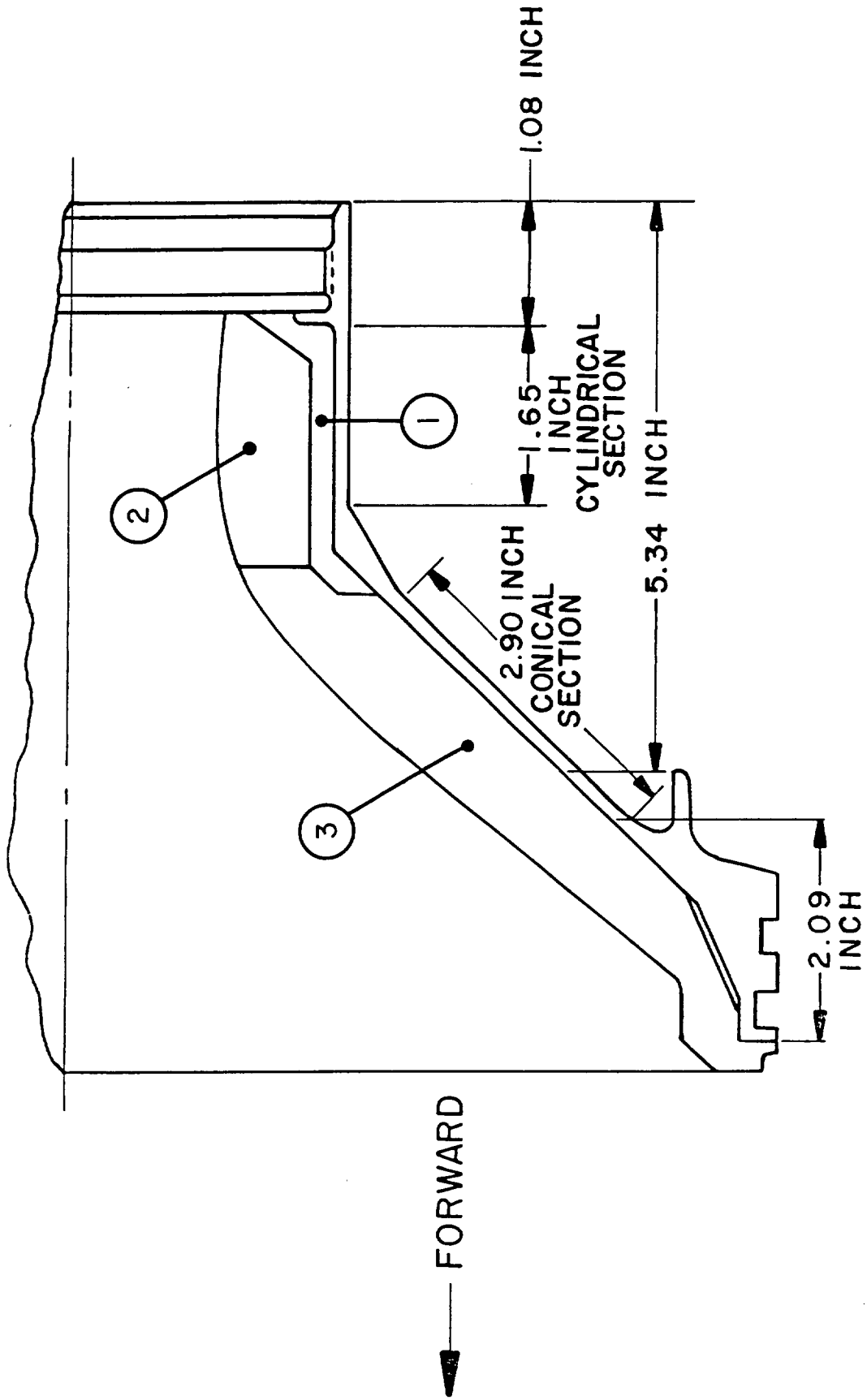


FIGURE 1. ULTRASONIC BOND INSPECTION

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3.6.2 Insulation. The temperature of the external surface of the shell shall be less than 285°C (550°F) in the conical section and 425°C (800°F) in the cylindrical section for the initial 150 seconds after motor ignition.

3.6.3 Component integrity (M103). There shall be no evidence of cracking of any ablative component (see 6.6.4) or motor Lot Acceptance Test (LAT) failure attributable to closure performance, e.g., burnthrough (see 6.6.4) during motor firing when the closure is assembled to a MK 56 rocket motor and tested in accordance with 4.7.5.

3.7 Workmanship. The closure assemblies shall be of uniform quality and shall be manufactured in accordance with standard manufacturing procedures of the aerospace industry. The requirements for workmanship shall be as specified in Dwgs 5546252 and 2841603 and the following:

3.7.1 Metal defects. The closure shell shall be free of cracks, splits, cold shuts, inclusions, porosity or any similar defect. There shall be no burrs which might interfere with the function of the closure or which might be injurious to personnel handling the item.

* 3.7.2 Molded components (M104). The molded components shall comply with the requirements of 3.7.2.1 and 3.7.2.2.

* 3.7.2.1 Find No. 1 insulator. The visible and exposed surfaces (in the case of a sectioned component) of the find No. 1 insulator shall be free of cracks and delaminations. There shall be no inclusions or voids larger than 0.06 X 0.06 X 0.125 inch. Inclusions shall be a minimum of 1.0 inch apart. There shall not be more than 8 inclusions in each molded insulator. Acceptable features include ripple surface, shrink-marks, resin voids, surface discoloration, weld-or knit-lines, surface crazing and no more than 50 pits.

* 3.7.2.2 Find No. 3 insulator. The visible and exposed surfaces (in the case of a sectioned component) of the find No. 3 insulator shall conform to the following:

a. No cracks are acceptable.

b. Voids whose greatest dimension exceeds 0.06 X 0.06 X 0.125 inch are not acceptable. No more than 15 voids nor more than two (2) surface voids within 4 inches of one another are acceptable.

c. Delaminations exceeding 0.75 inch are not acceptable. The total accumulative length of all delaminations in each component shall not exceed 2.25 inches. The minimum distance between any two delaminations shall not be less than three times the greatest dimension of the larger delamination.

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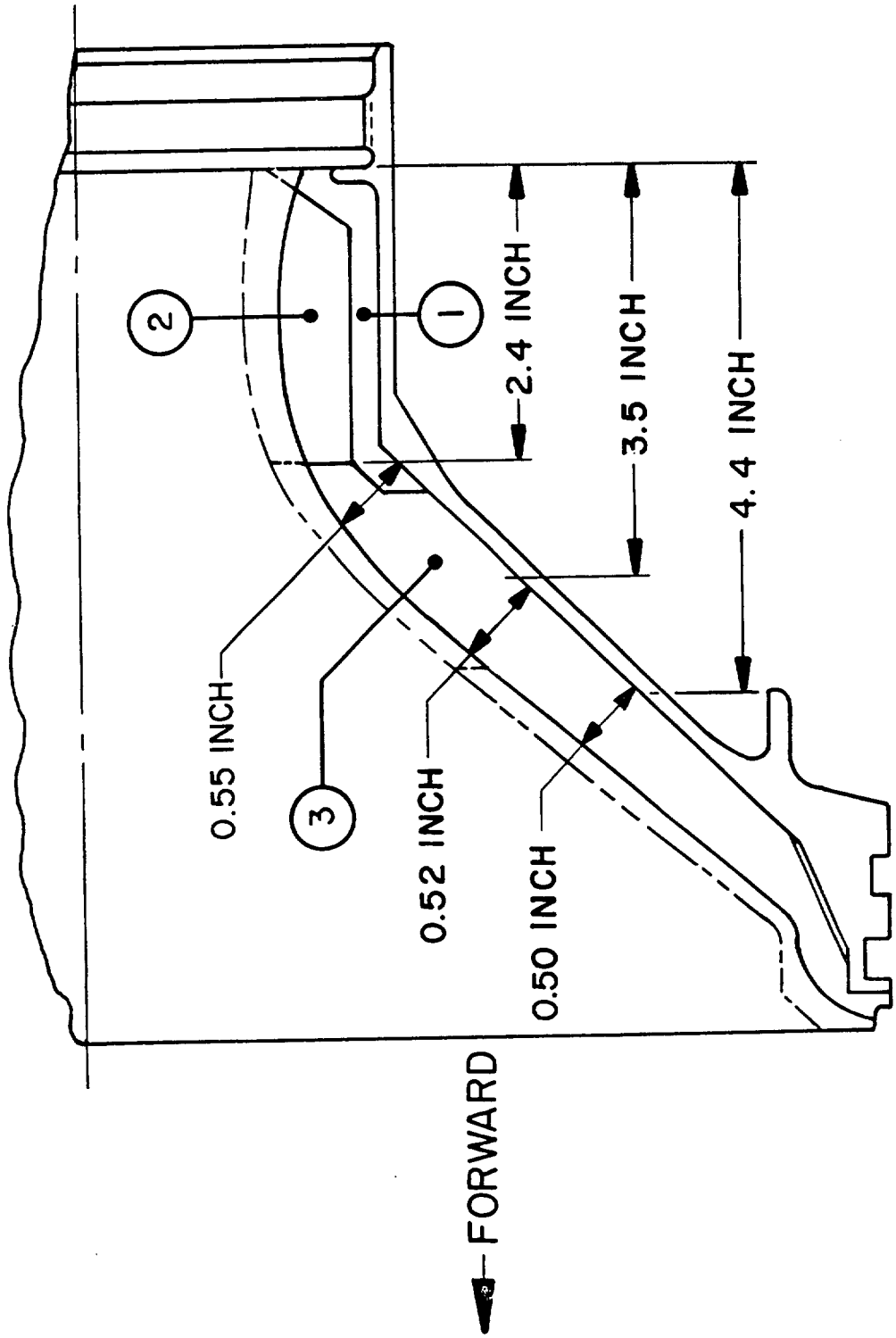


FIGURE 2. POST FIRE EROSION CONTOUR (MINIMUM)

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d. A maximum of 3 pieces of adhesive scrim is allowable in each component. Those pieces of scrim which are stringlike, less than 0.04 inch nominal diameter, shall not exceed 1.13 inches in length. Adhesive scrim chips whose maximum dimension exceeds 0.50 inch are not acceptable. The minimum distance between any two peices of scrim shall not be less than the greatest dimension of the larger piece.

e. A maximum of 6 pieces of cured silica-phenolic molding compound is allowable in each component. Those pieces of molding compound which are stringlike, less than 0.04 inch nominal diameter, shall not exceed 1.13 inches in length. Molding compound chips whose maximum dimension exceeds 0.50 inch are not acceptable. The minimum distance between any two pieces of molding compound shall not be less than greatest dimension of the larger piece.

f. A maximum of 4 pieces of miscellaneous foreign inclusions including metal is allowable in each component. Wirelike pieces exceeding 0.5 inch in length are not acceptable. Other forms of inclusions (such as chips) whose maximum dimension exceed 0.25 inch are not acceptable. The minimum distance between any two miscellaneous foreign inclusions shall not be less than the greatest dimension of the larger piece.

g. A maximum of 10 dry-spots are allowable on each component providing the maximum dimension of any dry-spot does not exceed 0.20 inch and the maximum depth of any dry-spot does not exceed 0.015 inch.

Acceptable features include ripple surface, shrink-marks, resin voids, surface discoloration, weld-or knit-lines, surface crazing and no more than 50 pits.

3.7.3 Machined graphite (M105). The machined graphite throat, Dwg 554252, find No. 2, shall be free of cracks, or surface irregularities.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all dimensional inspections shall be performed under the following conditions:

a. Temperature: Room ambient 18 to 35°C (65 to 95°F)
b. Humidity: Room ambient to 95 percent relative, maximum

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TABLE II. First article inspection

Inspection	Method	Sample
Visual and Dimensional	4.7.1	
Components	4.7.1.1	All
Closure assembly	4.7.1.2	All
Alcohol wipe test	4.7.2	
Components	4.7.2.1	All
Closure Assembly	4.7.2.2	All
Ultrasonic Test	4.7.3	All
Destructive Test	4.7.4	
Components	4.7.4.1 thru 4.7.4.4	All
Closure assembly	4.7.4.5	1, minimum
Performance	4.7.5	3, minimum

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4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

c. Periodic production inspection (see 4.6)

4.4 First article inspection. Unless otherwise specified in the contract (see 6.2) a first article sample of 5 closure assemblies and 3 sets of production components, find nos. 1, 2 and 3 of Dwg 5546252 (see 6.6.2) manufactured in accordance with 3.1, shall be submitted to an activity designated by the procuring activity (see 6.2) for first article inspection. The first article samples shall be inspected in accordance with Table II. Acceptance of the first article shall be based on no defects in the sample. Further production of closures by the contractor prior to approval of the preproduction sample shall be at the contractor's risk. First article samples shall not be included as a part of the quantity specified for delivery in the contract (see 6.2). When specified in the contract (see 6.2), a first article inspection report shall be prepared and submitted to the procuring activity.

4.5 Quality conformance inspection. The quality conformance inspection shall consist of the examinations and tests in 4.7.1 thru 4.7.4 performed on samples selected as specified in 4.5.1 for closure assemblies and components, and in accordance with WS 4503 for assembled closures.

4.5.1 Sampling for quality conformance inspection.

4.5.1.1 Lot. A closure lot shall consist of a maximum of 50 closure assemblies, assembled within a three week period by one manufacturer, at one plant, using the procedure detailed in the manufacturing process specification per 3.4.2. A lot of any component shall consist of those components manufactured in one continuous production run under essentially identical conditions, from the same lot of molding compound or graphite stock.

4.5.1.2 Sampling for visual examination. Each closure assembly shall be subjected to the examination of 4.7.1 for critical and major characteristics; sampling for minor characteristics shall be in accordance with inspection level II of MIL-STD-105.

4.5.1.3 Sampling for alcohol wipe test. Each component and each closure assembly shall be subjected to the alcohol wipe test of 4.7.2.

4.5.1.4 Sampling for ultrasonic test. Each closure assembly shall be subjected to the ultrasonic test of 4.7.3.

4.5.1.5 Sampling for destructive tests and examination. One molded component from each 100 or less comprising a molded component lot shall be subjected to the tests and examination of 4.7.4.1 through 4.7.4.4.

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4.5.1.6 Sampling for LAT firings. Sampling of closure shall be as specified for loaded rocket motors in WS 4503.

* 4.6 Periodic production inspection. One closure assembly from a maximum of 200 closure assemblies manufactured by each supplier, one closure from the first lot produced following a failure, and one closure assembly from the first lot following a six month or longer interruption in production, that have passed the quality conformance inspection of 4.5, shall be submitted to an activity designated by the procuring activity (see 6.2) for periodic production inspection. Periodic production samples shall be subjected to the test and examinations of 4.7.5. Acceptance of the periodic production sample shall be based on no defects. Disposition of lots represented by failed closures, including the preceding three lots, shall be as specified by the procuring activity (see 6.2). When specified in the contract (see 6.2), periodic production inspection of closure may be combined with lot acceptance testing (LAT) of motors. Periodic production samples shall not be included as part of the quantity specified for delivery in the contract (see 6.2). When specified in the contract, a periodic production inspection report shall be prepared and submitted to the procuring activity (see 6.2).

4.7 Examination and tests.

4.7.1 Visual and dimensional examination.

4.7.1.1 Components. Prior to assembly, all molded component, graphite throat, and shell samples (see 4.5.1.2) shall be visually examined to verify conformance with 3.7. The acceptable quality level (AQL) for minor characteristics shall be 4.0 percent defective. No defects are acceptable for major and critical characteristics.

4.7.1.2 Closure assembly. After assembly, all visible surfaces of each closure shall be visually and dimensionally examined to verify conformance with Dwg 5546252 and the requirements of 3.7.

4.7.2 Alcohol wipe test. The alcohol wipe test shall be performed in accordanc with the following to verify conformance with 3.7.2 and 3.7.3.

a. Thoroughly wet the part to be tested by application by brush, sponge or swab with alcohol conforming to ASTM D 304, O-M-232, grade A, or TT-I-735.

b. Observe the parts as the alcohol evaporates. The presence of cracks open to the surface will be indicated by a wet line, which will usually remain after the neighboring areas appear dry. Wide or deep cracks will remain wet longer than narrow or shallow cracks.

4.7.2.1 Components. Prior to assembly, the molded components and the graphite throat insert shall be alcohol wipe tested to verify conformance with 3.7.2 and 3.7.3. All surfaces of each component shall be tested. The presence of any defect specified in 3.7.2 or 3.7.3 shall result in rejection of the component.

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4.7.2.2 Closure assembly. After assembly, all visible interior surfaces of the closure shall be alcohol wipe tested to verify conformance with 3.7.2 and 3.7.3. The presence of any defect specified in 3.7.2 or 3.7.3 shall result in rejection of the closure.

4.7.3 Ultrasonic test. The insulation-to-steel bond of each assembled closure shall be ultrasonically tested in accordance with ASTM E 114 over the cylindrical and the conical sections specified in Figure I to verify conformance with 3.4.1. Any closure not meeting the minimum requirements of 3.4.1 shall be rejected. Test personnel qualification and certification shall be in accordance with MIL-STD-410.

4.7.4 Destructive tests and examination. The samples shall be subjected to the tests and examination of 4.7.4.1 through 4.7.4.5 as specified in 4.5.1.5 and TABLE II. Failure of a sample to meet any requirement shall result in the rejection of the lot represented by that sample.

4.7.4.1 Subsurface examination. Each sample molded component shall be sectioned in a minimum of five places along planes perpendicular to the center line of the part. The parts shall then be visually examined to verify conformance to 3.7.2.

4.7.4.2 Acetone extraction test. Each sample molded component shall be subjected to an acetone extraction test in accordance with ASTM D 494. Failure to conform to the requirement of 3.5 shall result in rejection of the sample.

4.7.4.3 Specific gravity test. Each sample molded component shall be subjected to a specific gravity test in accordance with ASTM D 792. Failure to conform to the requirement of 3.5 shall result in rejection of the sample.

4.7.4.4 Barcol hardness test. Each sample molded component shall be subjected a Barcol hardness test in accordance with ASTM D 2583. Failure to conform to the requirement of 3.5 shall result in rejection of the sample.

4.7.4.5 Bondline examination. Each sample closure shall be sectioned in a minimum of two places along planes passing through the centerline of the part. The parts shall then be visually examined to verify conformance to 3.4.1.

* 4.7.5 Performance. Sample closures shall be assembled to loaded MK 56 rocket motors and subjected to a static test firing in accordance with WS 4503. For first article inspection one motor shall be conditioned and stabilized at low temperature and two motors at high temperature prior to firing. For periodic production inspection, the firing temperature will be either high or low temperature depending on the random selection of the LAT motor. For LAT firings, conditioning shall be as specified in WS 4503. Motors shall be fired within 45 minutes after removal from the conditioning oven. Thermocouples shall be attached to each first article and periodic production sample closure to verify, via time vs. temperature records, conformance with 3.6.2.

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4.7.5.1 First article and periodic production post-fire examination. Following static firing, each sample closure shall be disassembled from the rocket motor and examined in accordance with the following:

a. Erosion. The inside diameter of the throat shall be measured at the point of maximum restriction to verify the requirement of 3.6.1. A minimum of 4 measurements, approximately 45° from one another, shall be taken. The closure shall then be sectioned through a plane containing the motor centerline and dimensionally examined to verify the contour requirement of 3.6.1. If one of the throat diameter measurements or any contour measurement fails to conform to the applicable requirement of 3.6.1, the sample shall be considered defective.

b. Component integrity. Following static firing each sample closure shall be visually examined to verify conformance to 3.6.3. A closure containing any crack, identified by rounded edges as having occurred during firing, as opposed to the sharp edges of post firing cracks, in any ablative component shall be considered defective.

4.7.5.2 LAT post-firing examination. Following each LAT motor firing the assembled closure shall be examined for evidence of any burnthrough or hot-spot (see 6.6) not indicative of normal closure performance. Upon occurrence of a burnthrough or hot spot all closures from the closure lot containing the defective closure and all succeeding closure lots shall be restricted from use pending satisfactory resolution of cause. Disposition of the suspect closure lot and screening of succeeding lots shall be as approved by the procuring activity.

4.8 Packaging inspection. The packaging, packing and marking shall be inspected to verify conformance with the requirement of section 5.

5. PACKAGING

5.1 Packaging and packing. Unless otherwise specified in the contract (see 6.2), packaging and packing shall be commercial.

5.1.1 Commercial. Unless otherwise specified in the contract (see 6.2), packaging and packing of the closure assemblies shall be in accordance with standard commercial practice applicable to the type of material. The packaging and packing shall be of such construction and material that the contents will be adequately protected against loss or contamination. Containers shall conform to Uniform Freight Classification, National Motor Freight Classification or to rules of other carriers applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract (see 6.2), each container shall be marked in accordance with MIL-STD-129, with the serial number of the item.

6. NOTES

6.1 Intended use. The aft closure is intended to be used as a component of the Mark 56 Mod 2 and Mod 3 Dual Thrust Rocket Motor in the Standard Missile (MR) system.

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6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.

b. Quantity required for delivery (does not include first article samples and periodic production samples, see 4.4 and 4.6).

c. When a first article sample is required (see 3.1, 4.4, 6.3, and 6.4).

d. Inspection conditions if other than as specified (see 4.2).

e. Assigned activity for first article and periodic production inspection (see 4.4 and 4.6).

f. Instructions for disposition of lots represented by failed periodic production sample closures (see 4.6).

g. Whether periodic production inspection of closures may be combined with motor LAT's (see 4.6).

h. Packaging requirements if other than commercial (see 5.1).

i. Any special markings required (see 5.2).

j. That the requirements of MIL-Q-9858 apply (see 6.5).

* 6.2.2 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of 27.410-6 of the DoD FAR Supplement are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirements	Applicable DID
3.4.2	Manufacturing Process Specification	UDI-E-20138
4.4	First Article Inspection Report	DI-T-4902
4.6	Periodic Production Inspection Report	DI-T-5247

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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* 6.3 First article sample. When a first article sample is required, it shall be tested and approved under the appropriate provision of 52.209-3 and 52.209-4 of the FAR. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test and approval of the first article sample.

6.4 First article sample waiver. First article samples submitted and approved on a recent contract may be accepted by the procuring activity in lieu of an additional first article inspection with the exception that component dimensional verification samples shall be required for each contract. When the first article sample is waived (see 6.2.1(c)), the procurement document should contain a statement specifying that the standard of workmanship exhibited by the previously approved first article sample shall determine the minimum requirements of the current contract.

* 6.5 Quality program requirements. Closure assemblies in accordance with this specification are complex-critical items as defined in FAR 46.203, and a quality program per MIL-Q-9858 should therefore be specified in the contract via reference to FAR 52.246-11.

6.6 Definitions. The following definitions apply to the terms used in this specification.

6.6.1 Closure assembly. The steel shell with all insulating and ablative components installed, referred to herein as the closure.

6.6.2 Molded components. The find Nos. 1 and 3 components of Dwg 5546252.

6.6.3 Ablative components. The find Nos. 2 and 3 components of Dwg 5546252.

6.6.4 Burnthrough. The burning or melting of the ablative components and/or the steel shell so as to result in a pathway for the hot gases to reach the atmosphere, other than the normal path through the nozzle.

6.6.5 Porous area. An area in the adhesive bond that ultrasonic testing indicates is not fully bonded while the signal does not indicate a void or unbonded situation. A porous area is characterized as having numerous small air bubbles or a sponge effect.

6.6.6 Hot spot. Any area on the outer surface of the closure shell where the paint is blistered or burned thus indicating the presence of excessive heat not found on the rest of the closure surface.

* 6.6.7 Void. A spherical or ellipsoidal shaped cavity either on or below the component surface.

* 6.6.8 Foreign inclusion. A particle of a substance imbedded in the component which is foreign to its composition.

* 6.6.9 Ripple surface. A surface feature resulting from the difference in shrinkage between the resin binder and reinforcement.

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* 6.6.10 Shrink-mark. A dimple-like depression in the surface of a molding where it has retracted from the mold, and which has well-rounded edges. A shrink-mark generally occurs on one surface of a part when there is a boss, flange, rib, or other heavy section on the opposite surface. The shrink-mark may be caused by the difference in total shrinkage when there is a sudden change in section along the surface of the part.

* 6.6.11 Resin void. Appearance of multiple surface interruptions which conform to the pattern of the cloth weave. Usually due to an insufficient flow or shrinkage of the resin, which fails to fill all of the interstices of the fabric reinforcement. These defects occur only at the surface layer of the resin in contact with the mold.

* 6.6.12 Surface discoloration. An apparent nonuniformity of a material evidenced by the appearance of contrasting colors.

* 6.6.13 Weld-or knit-line. Mark on, or weakness in, a molded plastic, formed by the union of two or more streams of plastic flowing together, that do not result in a break or interruption of the molded surface.

* 6.6.14 Surface crazing. Fine cracks in the component that do not retain alcohol when subjected to the procedure of 4.7.2.

* 6.6.15 Pit. Small crater in the surface of the part with its width no more than 0.015 inch and a depth no more than 0.020 inch.

* 6.6.16 Delamination. A separation in the component surface which follows the grain flow of the cured molding compound and retains alcohol when subjected to the procedure of 4.7.2.

* 6.6.17 Crack. A separation in the component surface which is transverse to the grain flow of the cured molding compound and retains alcohol when subjected to the procedure of 4.7.2.

* 6.6.18 Dry-spot. A molded surface feature characterized by an incomplete surface film where the reinforcement fabric has not been wetted with resin. This definition is applicable only to find no. 3 insulator of Dwg 5546252. A dry-spot is usually a result of an agglomeration of dust composed of fine particles of the carbon-phenolic molding compound.

* 6.6.19 Adhesive scrim. Coarsely woven fabric strip of glass fibers coated on both sides with an adhesive. Usually used to join sections of the reinforcing fabric prior to the application of the impregnating resin.

6.7 Adhesive bonding. MIL-A-83377 provides a detailed summary of state-of-the-art adhesive bonding practices that should be considered when establishing the manufacturing process specification specified in 3.4.2.

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6.8 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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